The empirical recurrence for OEIS A234407: recovering a conjecture that is not written down

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Abstract

OEIS A234407 records "Empirical recurrence of order 52", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 343$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 52, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 52 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234407 is "Number of (n+1)X(2+1) 0..6 arrays with every 2X2 subblock having the sum of the absolute values of the edge differences equal to 12 and no adjacent elements equal.". It has offset 1 and begins

$$2540, 38236, 578080, 9552088, 160156724, 2732474648, 46838063500, 805631496848, \ldots$$

The entry states:

> Empirical recurrence of order 52 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:22:48, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 343$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 343$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 686$ exact terms of the model returns order 52 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{52} c_i \, a(n-i),$$

c_1	16	c_{14}	1541912922843	c_{27}	−10094140918145782178	c_{40}	−85077115428166604124
c_2	343	c_{15}	−51407941504309	c_{28}	−2983247144563180 7	c_{41}	8484172356156996844 2
c_3	−5407	c_{16}	−15942622484437	c_{29}	3224315315217535754 1	c_{42}	5378280252628245710 8
c_4	−46193	c_{17}	737978911236175	c_{30}	1865095390289233426	c_{43}	−26360096701631253048
c_5	756458	c_{18}	105379206422301	c_{31}	−7967130101091653654 0	c_{44}	−21907925782469404952
c_6	3398105	c_{19}	−8155423255985012	c_{32}	−10270789071172665814	c_{45}	3927435154261656032
c_7	−59804847	c_{20}	−311647172223557	c_{33}	15065379666030899703 8	c_{46}	5391531414607092160
c_8	−156279725	c_{21}	70008759690721462	c_{34}	3190986755500259034 5	c_{47}	108401158942182752
c_9	3043046950	c_{22}	−1370629770032941	c_{35}	−21458516603144121961 1	c_{48}	−708154814934092992
c_{10}	4813601223	c_{23}	−469286539060437243	c_{36}	−6498279721877295711 0	c_{49}	−113176559199023616
c_{11}	−107016666546	c_{24}	19020944311183363	c_{37}	2250376727793008154 85	c_{50}	343150954925598 72
c_{12}	−102861688571	c_{25}	2461488830432656799	c_{38}	900522056647296192 83	c_{51}	103302680205772 80
c_{13}	2719081101472	c_{26}	−77120431959995216	c_{39}	−1680532340179558758 73	c_{52}	647234300805120

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 52, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $52+4$ terms removed returns order 52 as well, so $\deg q_{\min} = 52 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $52+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 52 that this sequence satisfies — and that family turns out to have exactly one member.

References

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